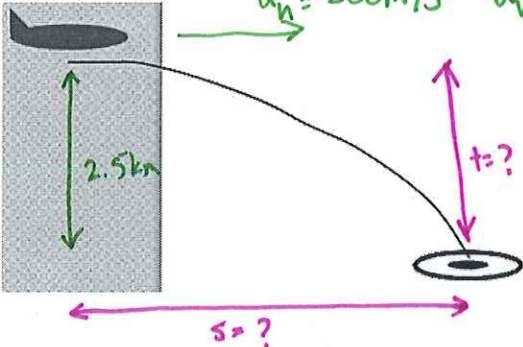


Class examples

1. A bomber is travelling horizontally at 300m/s at an altitude of 2.5 km towards its designated ground target. How far ahead should the target be when the bomb is dropped to hit the target?



$u_H = 300 \text{ m/s}$ $u_V = 0$

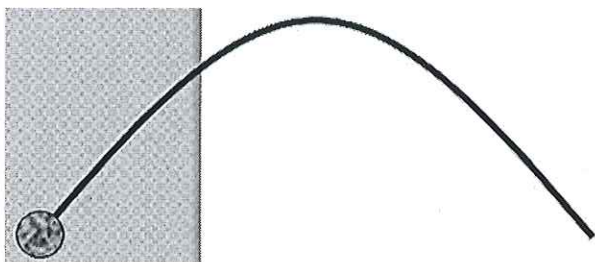
$v_H = \frac{s_H}{t} \Rightarrow s_H = v_H t$
 $s_V = u_V t + \frac{1}{2} a_V t^2$
 $s_V = \frac{1}{2} a_V t^2$

$t = \sqrt{\frac{2s_V}{a_V}} = \sqrt{\frac{2 \times (2.5)}{-9.8}} = 22.59 \text{ s}$

$s_H = (300) \times 22.59$
 $= \underline{6780 \text{ m}}$

Coordinate system diagram: + up, - down, - left, + right.

2. A soccer player kicks the ball with an initial velocity of 18.0 m/s at 35° above the horizontal.
 - a) Calculate the time the ball is in the air.
 - b) How far away will the ball land from the player's position?



at top $v_V = 0$
 $v_V = u_V + at$

a)



$u_V = u \sin \theta$
 $= 18 \sin(35) = 10.32 \text{ m/s}$

$v = u + at \Rightarrow t = \frac{v - u}{a} = \frac{0 - 10.32}{-9.8} = 1.053 \text{ s}$

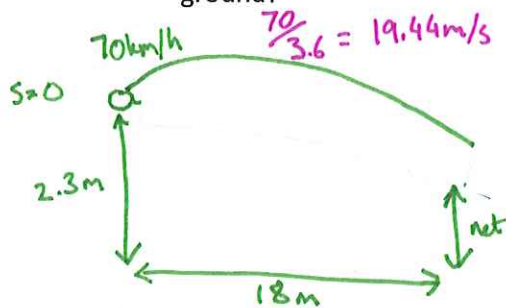
for the whole trip
 $1.053 \times 2 = \underline{2.11 \text{ s}}$

b)

$s_H = v_H \times t = 14.74 \times 2.11 = \underline{31.1 \text{ m}}$

$u_H = v_H = u \cos 35$
 $= 18 \cos 35$
 $= 14.74 \text{ m/s}$

3. During a tennis match, a player serves the ball by smacking it towards the opponent's half. When the racquet makes contact with the ball it is 2.3 m above the ground, and it is initially hit at a speed of 70 km/h 10.0° above the horizontal. The ball only just sails over the net which was 18.0 m away from the serving player. How high is the top of the net above the ground?



$$70 \frac{\text{km}}{\text{h}} = 19.44 \frac{\text{m}}{\text{s}}$$

$$u \sin \theta = 19.44 \sin 10 = 3.376 \frac{\text{m}}{\text{s}}$$

$$u \cos \theta = 19.44 \cos 10 = 19.14 \frac{\text{m}}{\text{s}}$$

$$v_h = \frac{s_h}{t} \quad t = \frac{s_h}{v_h} = \frac{18}{19.14} = 0.94025$$

$$s_v = u_v t + \frac{1}{2} a_v t^2$$

$$= (3.376)(0.9402) + \frac{1}{2} (-9.8)(0.9402)^2$$

$$= -1.153 \text{ m}$$

$$\therefore \text{net height} = 2.3 - 1.153$$

$$= \underline{1.14 \text{ m}}$$

Notes

A projectile is an object moving through the air where the only force acting on the object is its own weight force.

Equations:

$$v_{av} = \frac{s}{t} \quad s = ut + \frac{1}{2} at^2$$

$$v = u + at \quad a = \frac{v - u}{t}$$

$$v_{av} = \frac{v + u}{2} \quad v^2 = u^2 + 2as$$

If air resistance is ignored.

Horizontal velocity is constant.

$$a_h = 0 \quad s_h = v_h t$$

Vertical acceleration is constant

$$a_v = -9.8 \text{ m/s}^2$$

Effect of air resistance: lower max height
shorter horizontal displacement
non-symmetrical path (comes down steeper)